1.06 Complex Numbers: Modulus-Argument Form

Worked Solutions 1–54

1 a)

$$|z| = \sqrt{4^2 + (-2)^2} = \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5}$$

1 b)

$$|w| = \sqrt{(-1)^2 + (3)^2} = \sqrt{1 + 9} = \sqrt{10}$$

1 c)

$$|2w| = \sqrt{(-2)^2 + (6)^2} = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10}$$

1 d)

$$\frac{z}{w} = \frac{4 - 2i}{-1 + 3i} \times \frac{-1 - 3i}{-1 - 3i} = \frac{-4 + 2i - 12i - 6}{1 + 9} = -1 - i$$

$$\left|\frac{z}{w}\right| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{1 + 1} = \sqrt{2}$$

2 a)

$$\arg(1 + i) = \arctan\left(\frac{1}{1}\right) = \frac{\pi}{4}$$

2 b)

$$\arg(1 + \sqrt{3}i) = \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$$

2 c)

$$\arg(-2i) = -\frac{\pi}{2}$$

2 d)

$$\arg(-2 - 2i) = -\frac{\pi}{2} - \arctan\left(\frac{2}{2}\right) = -\frac{3\pi}{4}$$

2 e)

$$\arg(-3) = \pi$$

3 a)

$$3i$$

3 b)

$$\frac{1}{8} + \frac{\sqrt{3}}{8}i$$

3 c)

$$-\sqrt{3} + i$$

3 d)

$$\frac{7\sqrt{3}}{2} - \frac{7}{2}i$$

3 e)

$$-6 - 6\sqrt{3}i$$

3 f)

$$\frac{1}{4} - \frac{1}{4}i$$

4 a)

$$z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = 1 + \sqrt{3}i$$

4 b)

$$z = 1\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$$

4 c)

$$z = 4\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right) = -2 - 2\sqrt{3}i$$

5 a)

$$|3+3i| = \sqrt{3^2+3^2} = \sqrt{9+9} = \sqrt{18}$$
$$\arg(3+3i) = \arctan\left(\frac{3}{3}\right) = \frac{\pi}{4}$$
$$\therefore 3+3i = \sqrt{18}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = 3\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

5 b)

$$|2i| = \sqrt{0^2+2^2} = \sqrt{0+4} = 2$$
$$\arg(2i) = \frac{\pi}{2}$$
$$\therefore 2i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$$

5 c)

$$|-4| = \sqrt{0^2+(-4)^2} = \sqrt{0+16} = 4$$
$$\arg(-4) = \pi$$
$$\therefore -4 = 4(\cos\pi + i\sin\pi)$$

5 d)

$$|-1+\sqrt{3}i| = \sqrt{(-1)^2+(\sqrt{3})^2} = \sqrt{1+3} = 2$$
$$\arg(-1+\sqrt{3}i) = \frac{\pi}{2} + \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{2\pi}{3}$$
$$\therefore -1+\sqrt{3}i = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

5 e)

$$|2-2i| = \sqrt{(2)^2+(-2)^2} = \sqrt{4+4} = 2\sqrt{2}$$
$$\arg(2-2i) = -\arctan\left(\frac{2}{2}\right) = -\frac{\pi}{4}$$
$$\therefore 2-2i = 2\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$$

5 f)

$$|-\sqrt{3}-i| = \sqrt{(-\sqrt{3})^2+(-1)^2} = \sqrt{3+1} = 2$$
$$\arg(-\sqrt{3}-i) = -\frac{\pi}{2} - \arctan\left(\frac{\sqrt{3}}{1}\right) = -\frac{5\pi}{6}$$
$$\therefore -\sqrt{3}-i = 2\left(\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right)$$

5 g)

$$|-7+7i| = \sqrt{(-7)^2+7^2} = \sqrt{49+49} = \sqrt{98} = 7\sqrt{2}$$
$$\arg(-7+7i) = \frac{\pi}{2} + \arctan\left(\frac{7}{7}\right) = \frac{3\pi}{4}$$
$$\therefore -7+7i = 7\sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$$

5 h)

$$|3-\sqrt{3}i| = \sqrt{3^2+(-\sqrt{3})^2} = \sqrt{9+3} = \sqrt{12} = 2\sqrt{3}$$
$$\arg(3-\sqrt{3}i) = -\arctan\left(\frac{\sqrt{3}}{3}\right) = -\frac{\pi}{6}$$
$$\therefore 3-\sqrt{3}i = 2\sqrt{3}\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$$

6 a)

$$|1+5i| = \sqrt{1^2+5^2} = \sqrt{1+25} = \sqrt{26}$$
$$\arg(1+5i) = \arctan\left(\frac{5}{1}\right) = 1.37$$
$$\therefore 1+5i = \sqrt{26}(\cos 1.37 + i\sin 1.37)$$

6 b)

$$|-2+5i| = \sqrt{(-2)^2+5^2} = \sqrt{4+25} = \sqrt{29}$$
$$\arg(-2+5i) = \pi - \arctan\left(\frac{5}{2}\right) = 1.95$$
$$\therefore -2+5i = \sqrt{29}(\cos 1.95 + i\sin 1.95)$$

6 c)

$$|4-i| = \sqrt{4^2+(-1)^2} = \sqrt{16+1} = \sqrt{17}$$
$$\arg(4-i) = -\arctan\left(\frac{1}{4}\right) = -0.245$$
$$\therefore 4-i = \sqrt{17}(\cos(-0.245) + i\sin(-0.245))$$

6 d)

$$|-3-7i| = \sqrt{(-3)^2+(-7)^2} = \sqrt{9+49} = \sqrt{58}$$
$$\arg(-3-7i) = -\pi + \arctan\left(\frac{7}{3}\right) = -1.98$$
$$\therefore -3-7i = \sqrt{58}(\cos(-1.98) + i\sin(-1.98))$$

6 e)

$$|\sqrt{2}+6i| = \sqrt{(\sqrt{2})^2+6^2} = \sqrt{2+36} = \sqrt{38}$$
$$\arg(\sqrt{2}+6i) = \arctan\left(\frac{6}{\sqrt{2}}\right) = 1.34$$
$$\therefore \sqrt{2}+6i = \sqrt{38}(\cos 1.34 + i\sin 1.34)$$

6 f)

$$\left|-1+\frac{3}{4}i\right| = \sqrt{(-1)^2+\left(\frac{3}{4}\right)^2} = \sqrt{1+\frac{9}{16}} = \frac{5}{4}$$
$$\arg\left(-1+\frac{3}{4}i\right) = \pi - \arctan\left(\frac{3}{4}\right) = 2.50$$
$$\therefore -1+\frac{3}{4}i = \frac{5}{4}(\cos 2.50 + i\sin 2.50)$$

6 g)

$$|2-\sqrt{3}i| = \sqrt{2^2+(-\sqrt{3})^2} = \sqrt{4+3} = \sqrt{7}$$
$$\arg(2-\sqrt{3}i) = -\arctan\left(\frac{\sqrt{3}}{2}\right) = -0.714$$
$$\therefore 2-\sqrt{3}i = \sqrt{7}(\cos(-0.714) + i\sin(-0.714))$$

6 h)

$$\left|-1-\frac{1}{2}i\right| = \sqrt{(-1)^2+\left(-\frac{1}{2}\right)^2} = \sqrt{1+\frac{1}{4}} = \frac{\sqrt{5}}{2}$$
$$\arg\left(-1-\frac{1}{2}i\right) = -\pi + \arctan\left(\frac{1}{2}\right) = -2.68$$
$$\therefore -1-\frac{1}{2}i = \frac{\sqrt{5}}{2}(\cos(-2.68) + i\sin(-2.68))$$

7 a)

The mistake is with the angle; the angle of $\frac{\pi}{6}$ gives a complex number with positive real and imaginary parts, but the number has both a negative real and imaginary part.

7 b)

$$\arg(-\sqrt{3}-i) = -\pi + \arctan\left(\frac{1}{\sqrt{3}}\right) = -\frac{5\pi}{6}$$
$$\therefore -\sqrt{3}-i = 2\left(\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right)$$

8 a)

$$\frac{5\pi}{2} = 2\pi + \frac{\pi}{2}$$
$$\therefore 4\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$$

8 b)

$-\sin\left(\frac{\pi}{3}\right) = \sin\left(-\frac{\pi}{3}\right)$ and $\cos\left(\frac{\pi}{3}\right) = \cos\left(-\frac{\pi}{3}\right)$

$$\therefore 2\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right)$$

8 c)

$$\frac{10\pi}{3} = 2\pi + \frac{4\pi}{3} \therefore -\frac{2\pi}{3}$$

$-\sin\left(-\frac{2\pi}{3}\right) = \sin\left(\frac{2\pi}{3}\right)$ and $\cos\left(-\frac{2\pi}{3}\right) = \cos\left(\frac{2\pi}{3}\right)$

$$\therefore 3\left(\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right)\right)$$

8 d)

$\frac{10\pi}{3} = 2\pi + \frac{4\pi}{3}$ giving $-5\left(\cos\left(\frac{4\pi}{3}\right) + i\sin\left(\frac{4\pi}{3}\right)\right)$

$r = -5$ becoming $r = 5$ gives: $5\left(\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right)$

9 a)

$$\frac{1}{1-3i} \times \frac{1+3i}{1+3i} = \frac{1}{10} + \frac{3}{10}i$$
$$\left|\frac{1}{10} + \frac{3}{10}i\right| = \sqrt{\left(\frac{1}{10}\right)^2 + \left(\frac{3}{10}\right)^2} = \frac{1}{\sqrt{10}}$$

and $\arg z = \arctan(3) = 1.25$

$$z = \frac{1}{\sqrt{10}}(\cos 1.25 + i\sin 1.25)$$

Or $z = \frac{\sqrt{10}}{10}(\cos 1.25 + i\sin 1.25)$

9 b)

$$\frac{1}{1+3i} \times \frac{1-3i}{1-3i} = \frac{1}{10} - \frac{3}{10}i$$
$$\left|\frac{1}{10} - \frac{3}{10}i\right| = \sqrt{\left(\frac{1}{10}\right)^2 + \left(-\frac{3}{10}\right)^2} = \frac{1}{\sqrt{10}}$$

and $\arg z = \arctan(-3) = -1.25$

$$z = \frac{1}{\sqrt{10}}(\cos(-1.25) + i\sin(-1.25))$$

or $z = \frac{\sqrt{10}}{10}(\cos(-1.25) + i\sin(-1.25))$

9 c)

$$\frac{5}{2-3i} \times \frac{2+3i}{2+3i} = \frac{10+15i}{4+9} = \frac{10}{13} + \frac{15}{13}i$$

$$\left|\frac{10}{13} + \frac{15}{13}i\right| = \sqrt{\left(\frac{10}{13}\right)^2 + \left(\frac{15}{13}\right)^2} = \frac{5\sqrt{13}}{13}$$

and $\arg z = \arctan\left(\frac{15}{10}\right) = 0.983$

$$z = \frac{5\sqrt{13}}{13}(\cos(0.983) + i\sin(0.983))$$

9 d)

$$\frac{-7+i}{7-2i} \times \frac{7+2i}{7+2i} = \frac{-51-7i}{49+4} = -\frac{51}{53} - \frac{7}{53}i$$

$$\left|-\frac{51}{53} - \frac{7}{53}i\right| = \sqrt{\left(-\frac{51}{53}\right)^2 + \left(-\frac{7}{53}\right)^2} = \frac{5\sqrt{106}}{53}$$

and $\arg z = -\pi + \arctan\left(\frac{7}{51}\right) = -3.01$

$$z = \frac{5\sqrt{106}}{53}(\cos(-3.01) + i\sin(-3.01))$$

10)

$$3\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right)$$

11 a)

$$|4-5i| = \sqrt{4^2 + (-5)^2} = \sqrt{41}$$

and $\arg z = -\arctan\left(\frac{5}{4}\right) = -0.896$

$$z = \sqrt{41}(\cos(-0.896) + i\sin(-0.896))$$

11 b)

$$z^* = \sqrt{41}(\cos(0.896) + i\sin(0.896))$$

12 a)

$z^2 - 2z + 2 = 0$ gives $(z-1)^2 + 1 = 0 \Rightarrow z = 1 \pm i$

$$|1 \pm i| = \sqrt{(1)^2 + (\pm 1)^2} = \sqrt{2}$$

$$\arg(1+i) = \arctan(1) = \frac{\pi}{4}$$

$$\arg(1-i) = -\arctan(1) = -\frac{\pi}{4}$$

$z = \sqrt{2}\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right)$ or $z = \sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$

12 b)

$9z^2 - 3z + 1 = 0$ gives $9\left(z - \frac{1}{6}\right)^2 + \frac{3}{4} = 0 \Rightarrow z = \frac{1}{6} \pm \frac{\sqrt{3}}{6}i$

$$\left|\frac{1}{6} \pm \frac{\sqrt{3}}{6}i\right| = \sqrt{\left(\frac{1}{6}\right)^2 + \left(\pm\frac{\sqrt{3}}{6}\right)^2} = \frac{1}{3}$$

$$\arg\left(\frac{1}{6} + \frac{\sqrt{3}}{6}i\right) = \arctan(\sqrt{3}) = \frac{\pi}{3}$$

$$\arg\left(\frac{1}{6} - \frac{\sqrt{3}}{6}i\right) = -\arctan(\sqrt{3}) = -\frac{\pi}{3}$$

$z = \frac{1}{3}\left(\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right)$ or $z = \frac{1}{3}\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right)$

12 c)

$z^2 + 8z + 64 = 0$ gives $(z+4)^2 + 48 = 0 \Rightarrow z = -4 \pm 4\sqrt{3}i$

$$|-4 \pm 4\sqrt{3}i| = \sqrt{(-4)^2 + (\pm 4\sqrt{3})^2} = 8$$

$$\arg(-4 + 4\sqrt{3}i) = \pi - \arctan\left(\frac{4\sqrt{3}}{4}\right) = \frac{2\pi}{3}$$

$$\arg(-4 - 4\sqrt{3}i) = -\pi + \arctan\left(\frac{4\sqrt{3}}{4}\right) = -\frac{2\pi}{3}$$

$z = 8\left(\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right)\right)$ or $z = 8\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right)$

12 d)

$k^2z^2 - 2kz + 2 = 0$ gives $(kz-1)^2 + 1 = 0 \Rightarrow z = \frac{1}{k} \pm \frac{1}{k}i$

$$\left|\frac{1}{k} \pm \frac{1}{k}i\right| = \sqrt{\left(\frac{1}{k}\right)^2 + \left(\pm\frac{1}{k}\right)^2} = \frac{\sqrt{2}}{k}$$

$$\arg\left(\frac{2}{k} + \frac{2}{k}i\right) = \arctan(1) = \frac{\pi}{4}$$

$$\arg\left(\frac{2}{k} - \frac{2}{k}i\right) = -\arctan(1) = -\frac{\pi}{4}$$

$z = \frac{\sqrt{2}}{k}\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right)$ or $z = \frac{\sqrt{2}}{k}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$

13 a)

$$|2 + 3i| = \sqrt{2^2 + 3^2} = \sqrt{13}$$

$$\arg(2 + 3i) = \arctan\left(\frac{3}{2}\right) = 0.983$$

$$\therefore z_1 = 2 + 3i = \sqrt{13}(\cos 0.983 + i\sin 0.983)$$

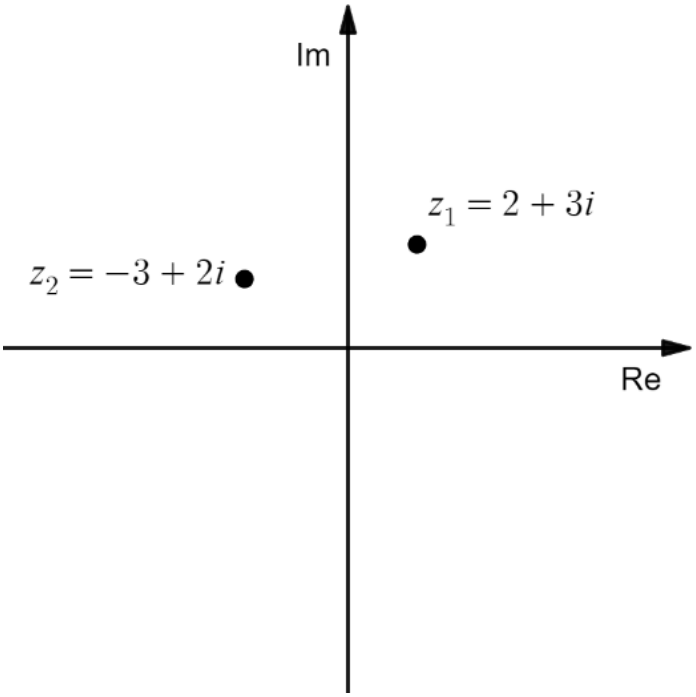
$$z_2 = -3 + 2i$$

$$|-3 + 2i| = \sqrt{(-3)^2 + 2^2} = \sqrt{13}$$

$$\arg(-3 + 2i) = \pi - \arctan\left(\frac{2}{3}\right) = 2.55$$

$\therefore z_1 = \sqrt{13}(\cos 0.983 + i\sin 0.983)$ and $z_2 = -3 + 2i = \sqrt{13}(\cos 2.55 + i\sin 2.55)$

13 b)



13 c)

Rotation $\frac{\pi}{2}$ anti-clockwise about the origin

14 a)

$$|1 + 2i| = \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$\arg(1 + 2i) = \arctan\left(\frac{2}{1}\right) = 1.11$$

$$\therefore z_1 = 1 + 2i = \sqrt{5}(\cos 1.11 + i \sin 1.11)$$

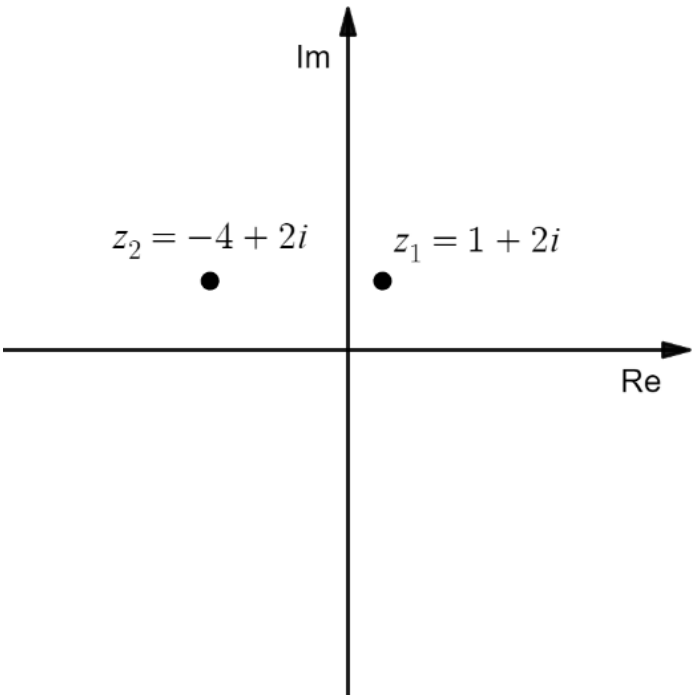
$$z_2 = 2i(1 + 2i) = -4 + 2i$$

$$|-4 + 2i| = \sqrt{(-4)^2 + 2^2} = 2\sqrt{5}$$

$$\arg(-4 + 2i) = \pi - \arctan\left(\frac{2}{4}\right) = 2.68$$

$\therefore z_1 = \sqrt{5}(\cos 1.11 + i \sin 1.11)$ and $z_2 = 2\sqrt{5}(\cos 2.68 + i \sin 2.68)$

14 b)



14 c)

$\pi - \arctan\left(\frac{2}{4}\right) - \arctan\left(\frac{2}{1}\right) = \frac{\pi}{2}$ and $2\sqrt{5} \div \sqrt{5} = 2$

Rotation $\frac{\pi}{2}$ anti-clockwise about the origin

Enlargement scale factor 2

15 a)

$$|3 + 2i| = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$\arg(3 + 2i) = \arctan\left(\frac{2}{3}\right) = 0.588$$

$$\therefore z_1 = 3 + 2i = \sqrt{13}(\cos 0.588 + i \sin 0.588)$$

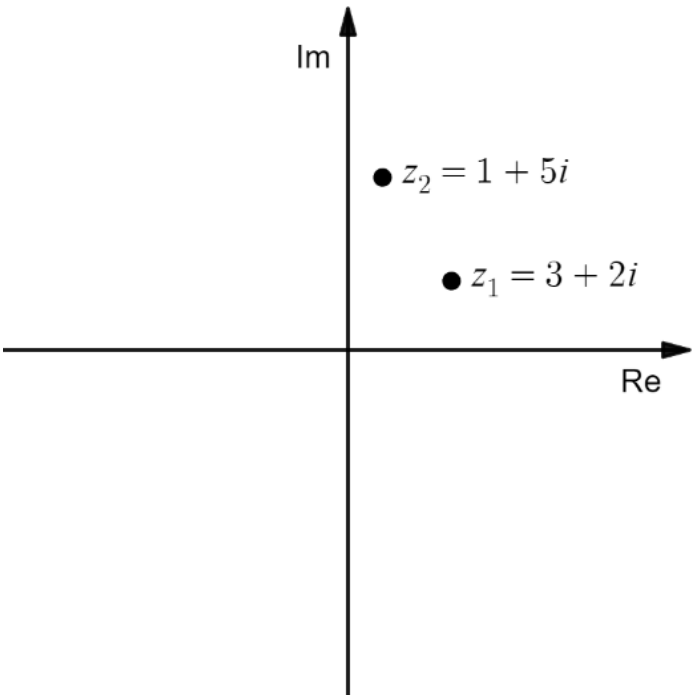
$$z_2 = (1 + i)(3 + 2i) = 1 + 5i$$

$$|1 + 5i| = \sqrt{(1)^2 + 5^2} = \sqrt{26}$$

$$\arg(1 + 5i) = \arctan\left(\frac{5}{1}\right) = 1.37$$

$\therefore z_1 = \sqrt{13}(\cos 0.588 + i \sin 0.588)$ and $z_2 = \sqrt{26}(\cos 1.37 + i \sin 1.37)$

15 b)



15 c)

$\arctan\left(\frac{5}{1}\right) - \arctan\left(\frac{2}{3}\right) = \frac{\pi}{4}$ and $\sqrt{26} \div \sqrt{13} = \sqrt{2}$

Rotation $\frac{\pi}{4}$ anti-clockwise about the origin

Enlargement scale factor $\sqrt{2}$

16 a)

$$3 + 2i + 1 - 5i = 4 - 3i \therefore |4 - 3i| = \sqrt{4^2 + (-3)^2} = 5$$

$$|3 + 2i| = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$|1 - 5i| = \sqrt{1^2 + (-5)^2} = \sqrt{26}$$

$\sqrt{13} + \sqrt{26} = 8.70$ (3 SF) and $5 < 8.70$

So $|z_1 + z_2| \leq |z_1| + |z_2|$

16 b)

By considering complex numbers as vectors: z_1, z_2 and $z_1 + z_2$ form a triangle

For any triangle the length of one side is always less than or equal to the sum of the lengths of the other two sides.

$$\therefore |z_1 + z_2| \leq |z_1| + |z_2|$$

17 a)

$$\arg(a - bi) = -\frac{\pi}{4}$$

17 b)

$$\arg(-a - bi) = -\frac{3\pi}{4}$$

17 c)

$$\arg(b + ai) = \frac{\pi}{4}$$

17 d)

$$\arg(b - ai) = -\frac{\pi}{4}$$

18 a)

$$\arg(a - bi) = \frac{\pi}{3}$$

18 b)

$$\arg(-a - bi) = \frac{2\pi}{3}$$

18 c)

$$\arg(b + ai) = \frac{5\pi}{6}$$

18 d)

$$\arg(b - ai) = -\frac{5\pi}{6}$$

19 a)

$$\theta = \arctan\left(\frac{2}{1}\right) - \arctan\left(\frac{3}{4}\right) = 0.464 \text{ (3 SF)}$$

Alternative Method:

$$\begin{aligned}\arg\left(\frac{z}{w}\right) = \arg\left(\frac{4+3i}{1+2i}\right) &= \arg(2 - i) \\ &= -\arctan\left(\frac{1}{2}\right) = -0.464 \text{ (3 SF)} \therefore 0.464 \text{ (3 SF)}\end{aligned}$$

19 b)

$$\theta = \pi - \arctan\left(\frac{6}{3}\right) - \arctan\left(\frac{2}{2}\right) = 1.25 \text{ (3 SF)}$$

Alternative Method:

$$\begin{aligned}\arg\left(\frac{z}{w}\right) &= \arg\left(\frac{2+2i}{-3+6i}\right) = \arg\left(\frac{2}{15} - \frac{6}{15}i\right) \\ &= -\arctan(3) = -1.24 \text{ (3 SF)} \therefore 1.24 \text{ (3 SF)}\end{aligned}$$

19 c)

$$\theta = \frac{\pi}{2} + \arctan\left(\frac{3}{4}\right) + \arctan\left(\frac{1}{1}\right) = 3.00 \text{ (3 SF)}$$

Alternative Method:

$$\begin{aligned}\arg\left(\frac{z}{w}\right) &= \arg\left(\frac{-4+3i}{1-i}\right) = \arg\left(-\frac{7}{2} - \frac{1}{2}i\right) \\ &= -\pi + \arctan\left(\frac{1}{7}\right) = -3.00 \text{ (3 SF)} \therefore 3.00 \text{ (3 SF)}\end{aligned}$$

20 a)

$|z_1| = \sqrt{2^2 + 1^2} = \sqrt{5}$ and $\arg z_1 = \arctan\left(\frac{1}{2}\right)$

$|z_2| = \sqrt{1^2 + 3^2} = \sqrt{10}$ and $\arg z_2 = \arctan(3)$

$$\arctan(3) - \arctan\left(\frac{1}{2}\right) = \frac{\pi}{4}$$

Area:

$$\frac{1}{2} \times \sqrt{5} \times \sqrt{10} \sin\left(\frac{\pi}{4}\right) = \frac{1}{2} \times \sqrt{5} \times \sqrt{10} \times \frac{\sqrt{2}}{2} = \frac{5}{2} \text{ units}^2$$

Alternative Method:

$$|z_1| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$|z_2| = \sqrt{1^2 + 3^2} = \sqrt{10}$$

$$\begin{aligned}\arg\left(\frac{z_1}{z_2}\right) &= \arg\left(\frac{2+i}{1+3i}\right) = \arg\left(\frac{1}{2} - \frac{1}{2}i\right) \\ &= -\arctan 1 = -\frac{\pi}{4} \text{ (3 SF)} \therefore \frac{\pi}{4} \text{ (3 SF)}\end{aligned}$$

Area:

$$\frac{1}{2} \times \sqrt{5} \times \sqrt{10} \sin\left(\frac{\pi}{4}\right) = \frac{1}{2} \times \sqrt{5} \times \sqrt{10} \times \frac{\sqrt{2}}{2} = \frac{5}{2} \text{ units}^2$$

20 b)

$|z_1| = \sqrt{1^2 + 2^2} = \sqrt{5}$ and $\arg z_1 = \arctan(2)$

$|z_2| = \sqrt{(-2)^2 + 4^2} = 2\sqrt{5}$ and $\arg z_2 = \pi - \arctan\left(\frac{4}{2}\right)$

$$\pi - \arctan\left(\frac{4}{2}\right) - \arctan(2) = 0.927\ldots$$

Area:

$$\frac{1}{2} \times \sqrt{5} \times 2\sqrt{5} \sin(0.927\ldots) = \frac{1}{2} \times \sqrt{5} \times 2\sqrt{5} \times \frac{4}{5} = 4 \text{ units}^2$$

Alternative Method:

$$|z_1| = \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$|z_2| = \sqrt{(-2)^2 + 4^2} = 2\sqrt{5}$$

$$\arg\left(\frac{z_1}{z_2}\right) = \arg\left(\frac{1+2i}{-2+4i}\right) = \arg\left(\frac{3}{10} - \frac{2}{5}i\right)$$

$$= -\arctan\left(\frac{4}{3}\right) = -0.927 \text{ (3 SF)} \therefore 0.927 \text{ (3 SF)}$$

Area:

$$\frac{1}{2} \times \sqrt{5} \times 2\sqrt{5} \sin(0.927\ldots) = \frac{1}{2} \times \sqrt{5} \times 2\sqrt{5} \times \frac{4}{5} = 4 \text{ units}^2$$

20 c)

$|z_1| = \sqrt{(-5)^2 + 3^2} = \sqrt{34}$ and $|z_2| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$

$$\arctan\left(\frac{3}{5}\right) + \frac{\pi}{2} + \arctan\left(\frac{2}{3}\right) = 2.699\ldots$$

Area:

$$\frac{1}{2} \times \sqrt{34} \times \sqrt{13} \sin(2.699\ldots) = \frac{1}{2} \times \sqrt{34} \times \sqrt{13} \times 0.428\ldots = \frac{9}{2} \text{ units}^2$$

Alternative Method:

$$|z_1| = \sqrt{(-5)^2 + 3^2} = \sqrt{34}$$

$$|z_2| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$$

$$\arg\left(\frac{z_1}{z_2}\right) = \arg\left(\frac{-5+3i}{2-3i}\right) = \arg\left(-\frac{19}{13} - \frac{9}{13}i\right)$$

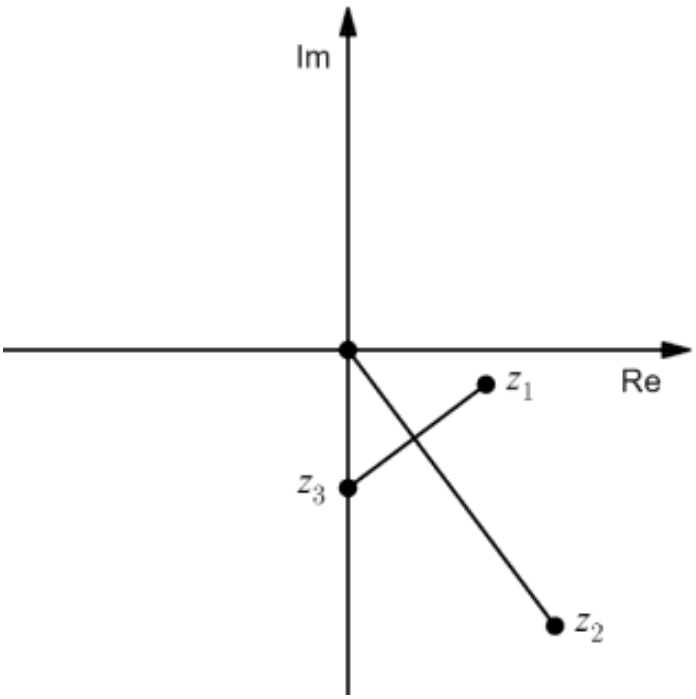
$$= -\pi + \arctan\left(\frac{9}{19}\right) = -2.699\ldots \text{ (3 SF)}$$

$\therefore 2.699\ldots$ (3 SF)

Area:

$$\frac{1}{2} \times \sqrt{34} \times \sqrt{13} \sin(2.699\ldots) = \frac{1}{2} \times \sqrt{34} \times \sqrt{13} \times 0.428\ldots = \frac{9}{2} \text{ units}^2$$

21)



Converting to cartesian:

$m_1 = -\frac{8}{6} = -\frac{4}{3}$ and $m_2 = \frac{-1--4}{4+0} = \frac{3}{4}$ so the diagonals are perpendicular so it is a kite.

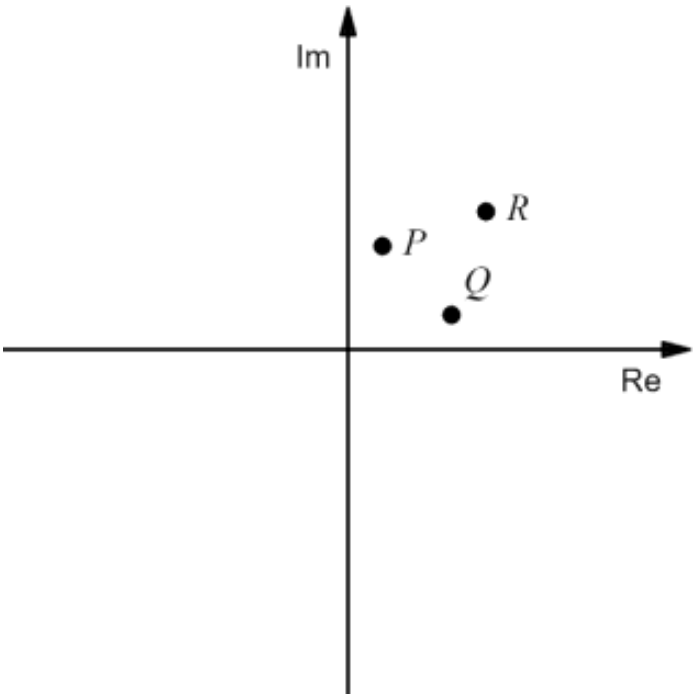
$$z_3 - z_1 = -4 - 3i \text{ so } |z_3 - z_1| = \sqrt{(-4)^2 + (-3)^2} = 5$$

$$|z_2| = \sqrt{6^2 + 8^2} = 10$$

$$\frac{1}{2} \times 5 \times 10 = 25$$

Therefore it has an area of 25 units^2

22 a)



22 b)

Let $\arg(z_1) = \theta_1$ and $\arg(z_2) = \theta_2$

The diagonal of the rhombus bisects the angle $\angle POQ$

$$\begin{aligned} \text{So } \arg(z_1 + z_2) &= \frac{1}{2}(\theta_1 - \theta_2) + \theta_2 \\ &= \frac{1}{2}\theta_1 + \frac{1}{2}\theta_2 \\ &= \frac{1}{2}\arg(z_1) + \frac{1}{2}\arg(z_2) \\ &= \frac{1}{2}(\arg z_1 + \arg z_2) \text{ as req.} \end{aligned}$$

23 a)

$3 \times 5 = 15$, $\frac{\pi}{4} + \frac{\pi}{3} = \frac{7\pi}{12}$

$$zw = 15\left(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right)$$

23 b)

$3 \div 5 = \frac{3}{5}$, $\frac{\pi}{4} - \frac{\pi}{3} = -\frac{\pi}{12}$

$$\frac{z}{w} = \frac{3}{5}\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right)$$

23 c)

$5 \div 3 = \frac{5}{3}$, $\frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$

$$\frac{w}{z} = \frac{5}{3}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$$

23 d)

$3 \times 3 = 9$, $\frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$

$$z^2 = 9\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$$

23 e)

$5^3 = 125$, $\frac{\pi}{3} + \frac{\pi}{3} + \frac{\pi}{3} = \pi$

$$w^3 = 125(\cos\pi + i\sin\pi)$$

24 a)

$2 \times 4 = 8$, $1.4 + 0.7 = 2.1$

$$zw = 8(\cos 2.1 + i\sin 2.1)$$

24 b)

$2 \div 4 = \frac{1}{2}$, $1.4 - 0.7 = 0.7$

$$\frac{z}{w} = \frac{1}{2}(\cos 0.7 + i\sin 0.7)$$

24 c)

$4 \div 2 = 2$, $0.7 - 1.4 = -0.7$

$$\frac{w}{z} = 2(\cos(-0.7) + i\sin(-0.7))$$

24 d)

$4^2 = 16$, $0.7 + 0.7 = 1.4$

$$w^2 = 16(\cos(1.4) + i\sin(1.4))$$

24 e)

$4^2 \times 2 = 32$, $0.7 + 0.7 + 1.4 = 2.8$

$$w^2 z = 32(\cos(2.8) + i\sin(2.8))$$

24 f)

$2^2 \div 4^2 = \frac{1}{4}$, $1.4 + 1.4 - 0.7 - 0.7 = 1.4$.

$$\frac{z^2}{w^2} = \frac{1}{4}(\cos 1.4 + i\sin 1.4)$$

25 a)

$7 \times 10 = 70$, $\frac{\pi}{6} + (-\frac{\pi}{3}) = -\frac{\pi}{6}$

$$zw = 70\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$$

25 b)

$7 \div 10 = \frac{7}{10}$, $\frac{\pi}{6} - (-\frac{\pi}{3}) = \frac{\pi}{2}$

$$\frac{z}{w} = \frac{7}{10}\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$$

25 c)

$10 \div 7 = \frac{10}{7}$, $(-\frac{\pi}{3}) - \frac{\pi}{6} = -\frac{\pi}{2}$

$$\frac{w}{z} = \frac{10}{7}\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right)$$

25 d)

$10^2 = 100$, $(-\frac{\pi}{3}) + (-\frac{\pi}{3}) = -\frac{2\pi}{3}$

$$2w^2 = 200\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right)$$

26)

$|1+i| = \sqrt{1^2+1^2} = \sqrt{2}$ and $\arg z = \arctan(1) = \frac{\pi}{4}$

$$z = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

$|\sqrt{3}+i| = \sqrt{(\sqrt{3})^2+1^2} = 2$ and $\arg z = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

$$w = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

$\sqrt{2}\times 2^2 = 4\sqrt{2}$, $\frac{\pi}{4}+\frac{\pi}{6}+\frac{\pi}{6} = \frac{7\pi}{12}$

$$zw^2 = 4\sqrt{2}\left(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right)$$

27 a)

$2\times 5 = 10$, $\frac{\pi}{4}+\frac{4\pi}{5} = \frac{21\pi}{20}$

But $\frac{21\pi}{20} > \pi$ so $-\frac{19\pi}{20}$

$$zw = 10\left(\cos\left(-\frac{19\pi}{20}\right) + i\sin\left(-\frac{19\pi}{20}\right)\right)$$

27 b)

$2\div 5 = \frac{2}{5}$, $\frac{\pi}{4}-\frac{4\pi}{5} = -\frac{11\pi}{20}$

$$\frac{z}{w} = \frac{2}{5}\left(\cos\left(-\frac{11\pi}{20}\right) + i\sin\left(-\frac{11\pi}{20}\right)\right)$$

27 c)

$5^3 = 125$, $\frac{4\pi}{5}+\frac{4\pi}{5}+\frac{4\pi}{5} = \frac{12\pi}{5}$

But $\frac{12\pi}{5} > \pi$ so $\frac{2\pi}{5}$

$$w^3 = 125\left(\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}\right)$$

27 d)

$$(1+i)z = (1+i)\times 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

Since $1+i = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ then:

$$\begin{aligned}(1+i)z &= 2\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)\\ &= 2\sqrt{2}\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)\end{aligned}$$

28 a)

$2 \times \frac{1}{2} = 1$, $-\frac{5\pi}{6} + \frac{3\pi}{4} = -\frac{\pi}{12}$

$$zw = 1\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right)$$

28 b)

$2 \div \frac{1}{2} = 4$, $-\frac{5\pi}{6} - \frac{3\pi}{4} = -\frac{19\pi}{12}$

But $-\frac{19\pi}{12} < -\pi$ so $\frac{5\pi}{12}$

$$\frac{z}{w} = 4\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$$

28 c)

$2^2 \times \left(\frac{1}{2}\right)^2 = 1$, $\left(-\frac{5\pi}{6}\right) + \left(-\frac{5\pi}{6}\right) + \frac{3\pi}{4} + \frac{3\pi}{4} = -\frac{\pi}{6}$

$$(zw)^2 = 1\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$$

28 d)

$$(1+\sqrt{3}i)w = (1+\sqrt{3}i) \times \frac{1}{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$$

Since $1 + \sqrt{3}i = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ then:

$$\begin{aligned}(1+\sqrt{3}i)w &= \frac{1}{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right) \times 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)\\ &= 1\left(\cos\frac{13\pi}{12} + i\sin\frac{13\pi}{12}\right)\\ &= 1\left(\cos\left(-\frac{11\pi}{12}\right) + i\sin\left(-\frac{11\pi}{12}\right)\right)\end{aligned}$$

29 a)

$|z| = \sqrt{(\sqrt{3})^2 + (\sqrt{3})^2} = \sqrt{6}$ and $\arg z = \arctan(1) = \frac{\pi}{4}$

$$z = \sqrt{6}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

29 b)

$|w| = \sqrt{0^2 + 4^2} = 4$ and $\arg w = \frac{\pi}{4}$

$$w = 4\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$$

29 c)

$\sqrt{6} \times 4^2 = 16\sqrt{6}$ and $\frac{\pi}{4} + \frac{\pi}{2} + \frac{\pi}{2} = \frac{5\pi}{4}$

But $\frac{5\pi}{4} > \pi$ so $-\frac{3\pi}{4}$

$$zw^2 = 16\sqrt{6}\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right)$$

30 a)

$$z = 3\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = -\frac{3}{2} + \frac{3\sqrt{3}}{2}i$$

30 b)

$$z^* = 3\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right) = -\frac{3}{2} - \frac{3\sqrt{3}}{2}i$$

30 c)

$$z^2 = 9\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)$$

$$z^2 = 9\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right) = -\frac{9}{2} - \frac{9\sqrt{3}}{2}i$$

30 d)

$3 \div 3 = 1$, $\frac{2\pi}{3} - \left(-\frac{2\pi}{3}\right) = \frac{4\pi}{3}$

$$\frac{z}{z^*} = 1\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

31 a)

$$\arg(z^2) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$$

31 b)

$$\arg(z^2) = -\frac{\pi}{3} - \frac{\pi}{3} = -\frac{2\pi}{3}$$

32 a)

$$\begin{aligned}\frac{z}{w} &= \frac{2}{3}\left(\cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right)\right) \\ &= \frac{2}{3}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right) = 0.644 + 0.173i\end{aligned}$$

32 b)

$$|1+i| = \sqrt{1^2+1^2} = \sqrt{2}$$
$$\arg(1+i) = \arctan(1) = \frac{\pi}{4}$$
$$\frac{z}{w} = \frac{\sqrt{2}}{2}\left(\cos\left(\frac{\pi}{4}-\frac{\pi}{6}\right)+i\sin\left(\frac{\pi}{4}-\frac{\pi}{6}\right)\right)$$
$$= \frac{\sqrt{2}}{2}\left(\cos\frac{\pi}{12}+i\sin\frac{\pi}{12}\right) = 0.683+0.183i$$

33 a)

$$|5-2i| = \sqrt{5^2+(-2)^2} = \sqrt{29}$$
$$\frac{\sqrt{290}}{\sqrt{29}} = \sqrt{10} \therefore |w| = \sqrt{10}$$

33 b)

$$\arg(5-2i) = -\arctan\left(-\frac{2}{5}\right)$$
$$\arg(w) = 0.8 - \arctan\left(-\frac{2}{5}\right) \therefore \arg(w) = 1.18 \text{ (3 SF)}$$

34 a)

$|z_1| = \sqrt{\left(-\frac{\sqrt{3}}{2}\right)^2+\left(-\frac{3}{2}\right)^2} = \sqrt{3}$ and $\arg z = -\pi + \arctan\left(\frac{-\frac{3}{2}}{-\frac{\sqrt{3}}{2}}\right) = -\frac{2\pi}{3}$

$$z_1 = \sqrt{3}\left(\cos\left(-\frac{2\pi}{3}\right)+i\sin\left(-\frac{2\pi}{3}\right)\right)$$

34 b)

$$|z_1 z_2| = \sqrt{30} = \sqrt{3}|z_2|$$

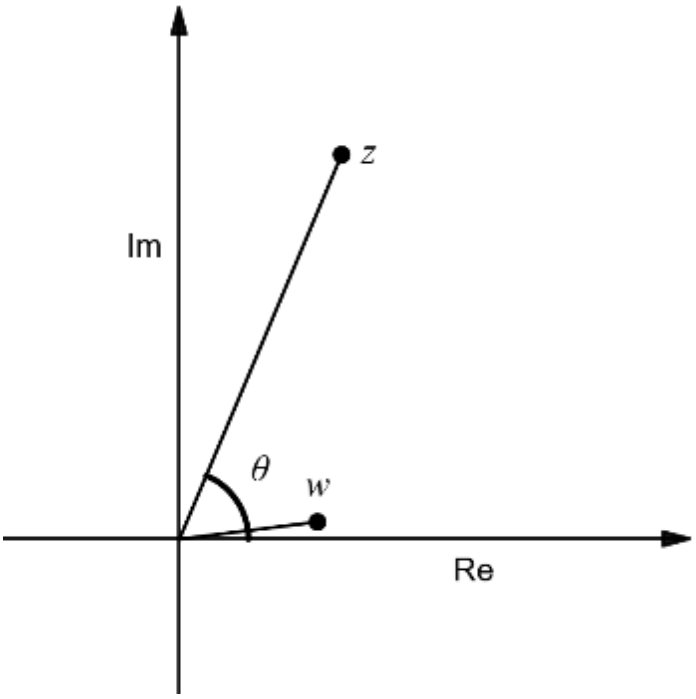
so $|z_2| = \sqrt{10}$

$\arg(z_1 z_2) = \frac{\pi}{4}$ so $-\frac{2\pi}{3} + \arg(z_2) = \frac{\pi}{4}$ so $\arg(z_2) = \frac{\pi}{4} + \frac{2\pi}{3} = \frac{11\pi}{12}$

$$\therefore z_2 = \sqrt{10}\left(\cos\left(\frac{11\pi}{12}\right)+i\sin\left(\frac{11\pi}{12}\right)\right)$$
$$= \left(-\frac{\sqrt{15}}{2}-\frac{\sqrt{5}}{2}\right)+\left(\frac{\sqrt{15}}{2}-\frac{\sqrt{5}}{2}\right)i$$

35)

Let $\arg z = \theta$



and so $\arg w = \theta - \frac{\pi}{3}$ since w has the smaller argument

$\arg\left(\frac{w}{z}\right) = \arg(w) - \arg(z) = \theta - \frac{\pi}{3} - \theta = -\frac{\pi}{3}$ and $\left|\frac{w}{z}\right| = \frac{|w|}{|z|} = \frac{|3z|}{|z|} = 3$

$$\frac{w}{z} = 3\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right) = \frac{3}{2} - \frac{3\sqrt{3}}{2}i$$

36 a)

$|z| = \sqrt{(-k)^2 + k^2} = \sqrt{2k^2} = \sqrt{2}k$ and $\arg z = \pi - \arctan(1) = \frac{3\pi}{4}$

$$z = \sqrt{2}k\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$$

36 b)

$$zw = \sqrt{2}k \times 2\sqrt{2}(\cos\pi + i\sin\pi)$$
$$-20 = 4k\cos\pi \Rightarrow k = 5$$

37)

$$iz^3 = 2^3 i\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$$
$$= 8i\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$$

Since $i = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$ then:

$$\begin{aligned}
iz^2 &= 8\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right) \\
&= 8\Bigg(\cos\frac{\pi}{2}\cos\frac{3\pi}{4} - \sin\frac{\pi}{2}\sin\frac{3\pi}{4} \\
&\qquad + i\sin\frac{\pi}{2}\cos\frac{3\pi}{4} + i\cos\frac{\pi}{2}\sin\frac{3\pi}{4}\Bigg) \\
&= 8\left(\cos\left(\frac{\pi}{2} + \frac{3\pi}{4}\right) + i\sin\left(\frac{\pi}{2} + \frac{3\pi}{4}\right)\right) \\
&= 8\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right) \\
&= 8\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right)
\end{aligned}$$

38 a)

Let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$

$$\begin{aligned}
z_1z_2 &= r_1(\cos\theta_1 + i\sin\theta_1) \times r_2(\cos\theta_2 + i\sin\theta_2) \\
&= r_1r_2(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) \\
&= r_1r_2(\cos\theta_1\cos\theta_2 + i\cos\theta_1\sin\theta_2 \\
&\qquad + i\sin\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) \\
&= r_1r_2((\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) \\
&\qquad + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)) \\
&= r_1r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))
\end{aligned}$$

So $|z_1z_2| = r_1r_2$ and $\arg(z_1z_2) = \theta_1 + \theta_2$

38 b)

Let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$

$$\begin{aligned}
\frac{z_1}{z_2} &= r_1(\cos\theta_1 + i\sin\theta_1) \div r_2(\cos\theta_2 + i\sin\theta_2) \\
&= \frac{r_1}{r_2} \times \frac{(\cos\theta_1 + i\sin\theta_1)}{(\cos\theta_2 + i\sin\theta_2)} \\
&= \frac{r_1}{r_2} \times \frac{(\cos\theta_1 + i\sin\theta_1)}{(\cos\theta_2 + i\sin\theta_2)} \times \frac{(\cos\theta_2 - i\sin\theta_2)}{(\cos\theta_2 - i\sin\theta_2)} \\
&= \frac{r_1}{r_2} \times \frac{(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 - i\sin\theta_2)}{(\cos^2\theta_2 + \sin^2\theta_2)} \\
&= \frac{r_1}{r_2}(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 - i\sin\theta_2) \\
&= \frac{r_1}{r_2}(\cos\theta_1\cos\theta_2 - i\cos\theta_1\sin\theta_2 \\
&\qquad + i\sin\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) \\
&= \frac{r_1}{r_2}((\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) \\
&\qquad + i(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2)) \\
&= \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2))
\end{aligned}$$

So $\arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2$

39)

$$\arg\left(\frac{1}{z}\right) = \arg 1 - \arg z = 0 - \arg z = -\arg z$$

Alternative Method:

Let $z = r(\cos\theta + i\sin\theta)$

$$\begin{aligned}\frac{1}{z} &= \frac{1}{r} \times \frac{1}{\cos\theta + i\sin\theta} \\ &= \frac{1}{r} \times \frac{1}{\cos\theta + i\sin\theta} \times \frac{\cos\theta - i\sin\theta}{\cos\theta - i\sin\theta} \\ &= \frac{1}{r} \times \frac{\cos\theta - i\sin\theta}{\cos^2\theta + \sin^2\theta} \\ &= \frac{1}{r} \times (\cos\theta - i\sin\theta)\end{aligned}$$

Since $\cos\theta = \cos(-\theta)$ and $-\sin\theta = \sin(-\theta)$

$$\frac{1}{z} = \frac{1}{r} \times (\cos(-\theta) - i\sin(-\theta))$$

So $\arg\left(\frac{1}{z}\right) = -\theta$

$$\therefore \arg\left(\frac{1}{z}\right) = -\arg z$$

40)

$$\begin{aligned}iz^2 &= 9i\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right) \\ &= 9i\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right)\end{aligned}$$

Since $i = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$ then:

$$\begin{aligned}iz^2 &= 9\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right) \\ &= 9\left(\cos\left(\frac{\pi}{2} - \frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3} + \frac{\pi}{2}\right)\right) \\ &= 9\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)\end{aligned}$$

41)

$|z^6| = 729$ so $|z| = 3$

$\arg(z^6) = 6\arg z$ so $\arg z = \frac{5\pi}{18}$

$$z = 3\left(\cos\frac{5\pi}{18} + i\sin\frac{5\pi}{18}\right)$$

So $a = 3\cos\frac{5\pi}{18}$ and $b = 3\sin\frac{5\pi}{18}$

42)

$$z = 2\sqrt{3} + 2i = 4\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

$$w = -\sqrt{2} + \sqrt{2}i = 2\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$$
$$r : 4 \to 2$$
$$\theta : \frac{3\pi}{4} - \frac{\pi}{6} = \frac{7\pi}{12}$$

Enlargement by scale factor $\frac{1}{2}$

Rotation by $\frac{7\pi}{12}$ anticlockwise about the origin.

43)

$$z = \sqrt{17}(\cos 1.3258\ldots + i\sin 1.3258\ldots)$$
$$w = \sqrt{13}(\cos 2.55359\ldots + i\sin 2.55359\ldots)$$
$$r : \sqrt{17} \to \sqrt{13}$$
$$\theta : 2.55359\ldots - 1.3258\ldots = 1.23 \text{ (3 SF)}$$

Enlargement scale factor $\frac{\sqrt{13}}{\sqrt{17}} = \frac{\sqrt{221}}{17} = 0.874$ (3 SF)

Rotate anti-clockwise 1.23 radians around the origin.

44)

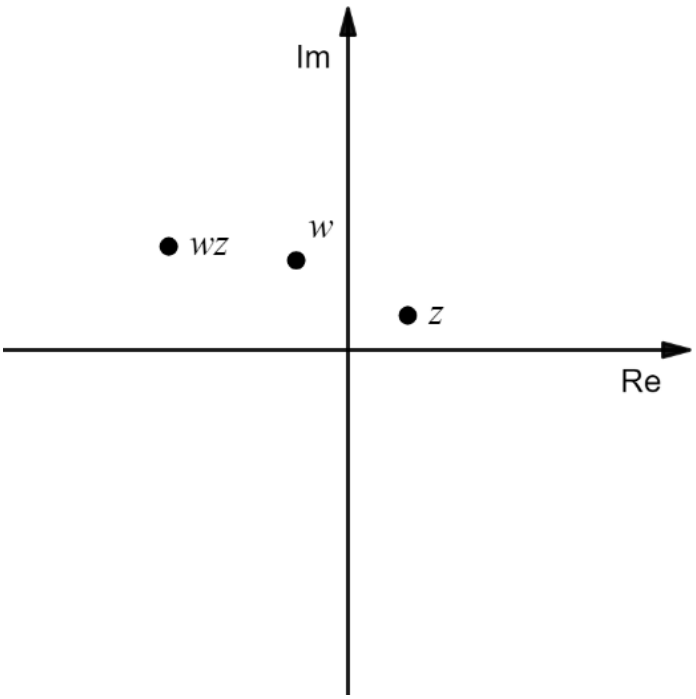
$$z = 2\sqrt{5}(\cos 0.4636\ldots + i\sin 0.4636\ldots)$$
$$w = \sqrt{26}(\cos -1.3734\ldots + i\sin -1.3734\ldots)$$
$$r : 2\sqrt{5} \to \sqrt{26}$$
$$\theta : -1.3734\ldots - 0.4636\ldots = -1.84 \text{ (3 SF)}$$

Stretch scale factor $\frac{\sqrt{26}}{2\sqrt{5}} = \frac{\sqrt{130}}{10} = 1.14$ (3 SF)

Rotate clockwise 1.84 radians around the origin.

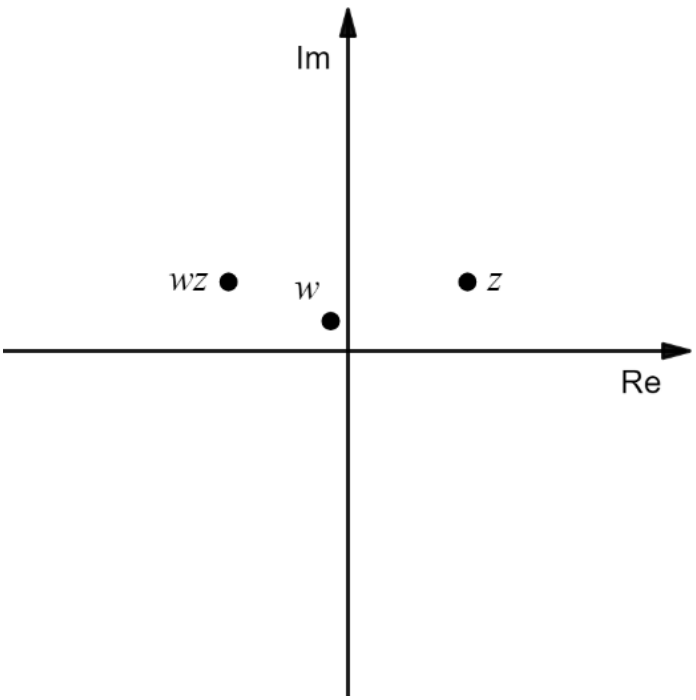
45 a)

$$z = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$
$$w = 3\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$
$$zw = 6\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$$



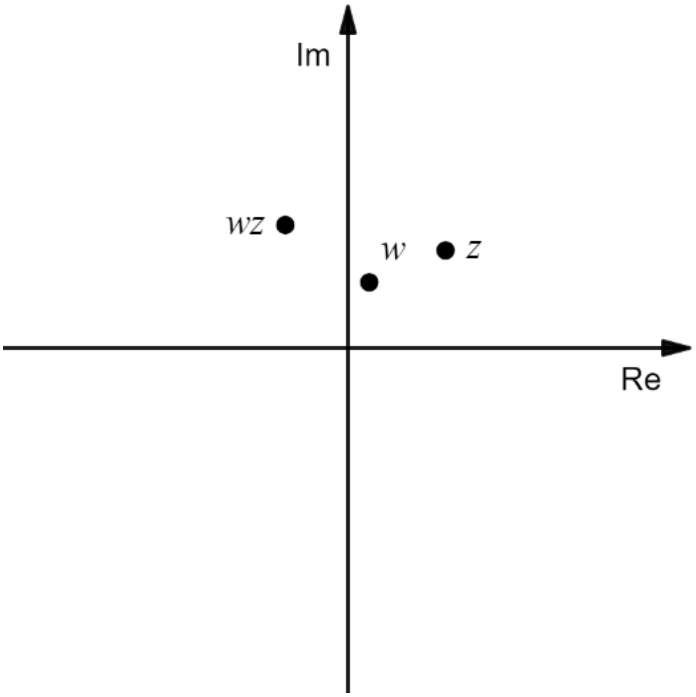
45 b)

$$z = 4\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$
$$w = 1\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$
$$zw = 4\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$$



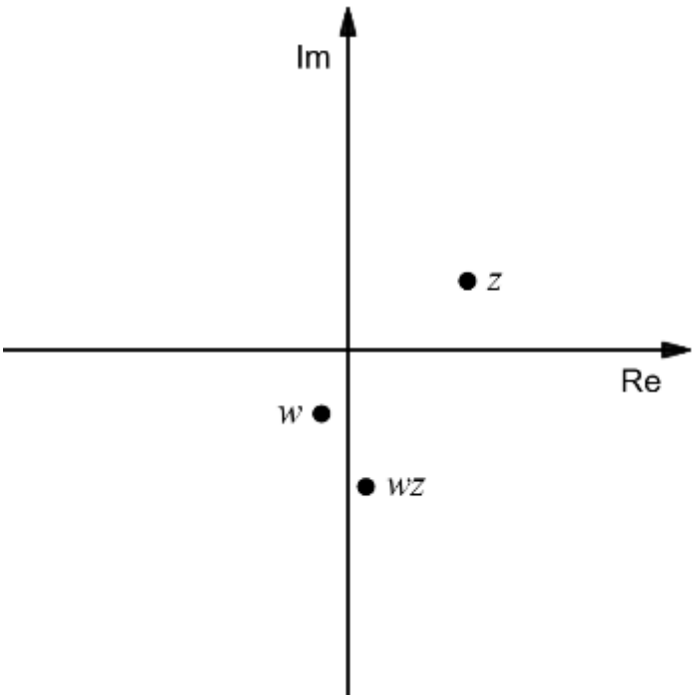
46 a)

$$z = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$
$$w = 1\left(\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}\right)$$
$$zw = 2\left(\cos\frac{13\pi}{20} + i\sin\frac{13\pi}{20}\right)$$



46 b)

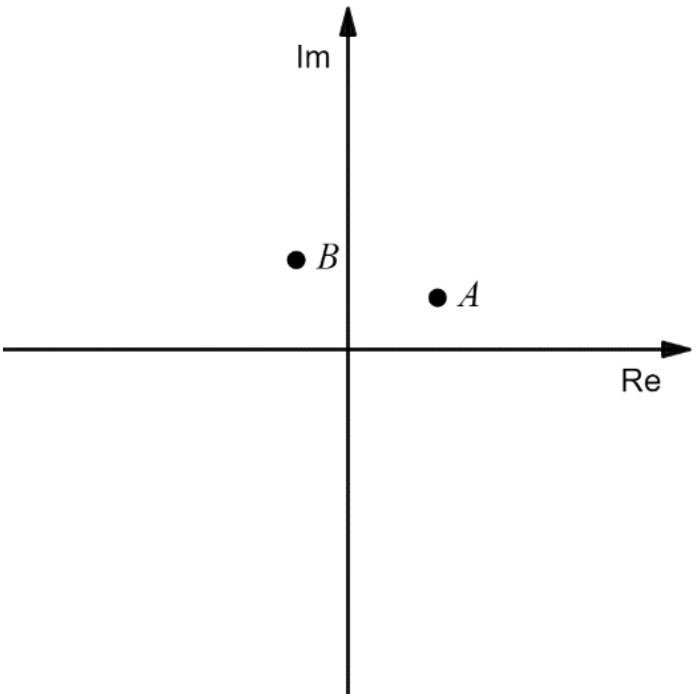
$$z = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$
$$w = 1\left(\cos\left(-\frac{5\pi}{8}\right) + i\sin\left(-\frac{5\pi}{8}\right)\right)$$
$$zw = 2\left(\cos\left(-\frac{11\pi}{24}\right) + i\sin\left(-\frac{11\pi}{24}\right)\right)$$



47 a)

$$z_1 = 3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right) = \frac{3\sqrt{3}}{2} + \frac{3}{2}i$$

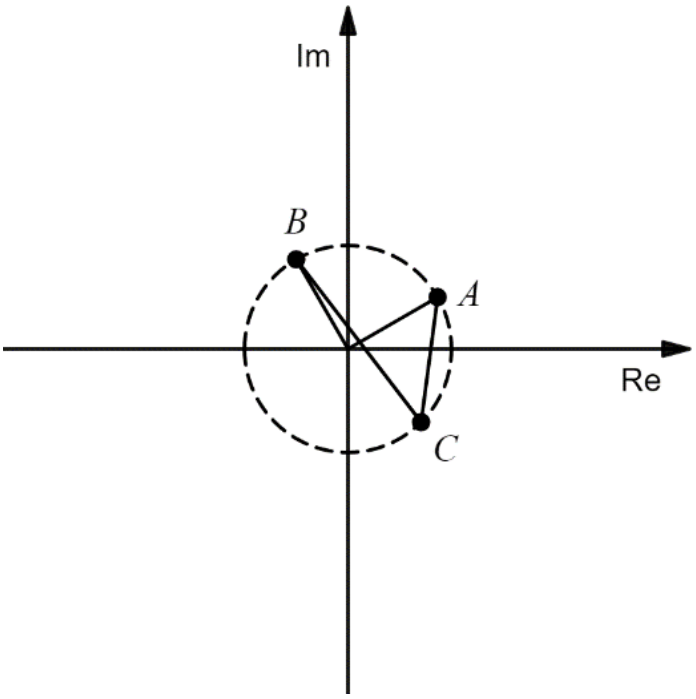
$$z_2 = 3\left(\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right)\right) = -\frac{3}{2} + \frac{3\sqrt{3}}{2}i$$



47 b)

All the points lie on a circle.

$$\frac{2\pi}{3} - \frac{\pi}{6} = \frac{\pi}{2}$$



Since the angle at the centre is twice the angle at the circumference, angle $\angle ACB = \frac{\pi}{4}$

48 a)

Let $z = a + bi$

$$|z|^2 = \left(\sqrt{a^2 + b^2}\right)^2$$
$$zz^* = (a + bi)(a - bi) = a^2 + b^2$$
$$\therefore |z|^2 = zz^*$$

48 b)

Let $z = a + bi$ and $w = c + di$

$$\begin{aligned}(z - w)^* &= (a + bi - (c + di))^* \\ &= (a - c + (b - d)i)^* \\ &= a - c - (b - d)i \\ &= a - bi - c + di \\ &= a - bi - (c - di) \\ &= z^* - w^* \text{ as req.}\end{aligned}$$

48 c)

$$\begin{aligned}|z - w|^2 &= (z - w)(z - w)^* = (z - w)(z^* - w^*) \\ &= zz^* - zw^* - wz^* + ww^*\end{aligned}$$

48 d)

$|z|^2 = zz^*$ so $zz^* = 2^2 = 4$

$|w|^2 = ww^*$ so $ww^* = 3^2 = 9$

$|z+w|^2 = (z+w)(z+w)^*$ so $4^2 = 16 = zz^* + wz^* + z^*w + ww^*$

$16 = 4 + wz^* + z^*w + 9$ so $wz^* + z^*w = 3$

$$\begin{aligned} |z-w|^2 &= zz^* - zw^* - wz^* + ww^* \\ &= 4 - 3 + 9 = 10 \end{aligned}$$

$$\therefore |z-w| = \sqrt{10}$$

49)

$$\begin{aligned} &|z+w|^2 + |z-w|^2 \\ &= (z+w)(z+w)^* + (z-w)(z-w)^* \\ &= zz^* + zw^* + z^*w + ww^* + zz^* - zw^* - z^*w + ww^* \\ &= 2zz^* + 2ww^* = 2|z|^2 + 2|w|^2 \text{ as req.} \end{aligned}$$

Alternative Method:

Let $z = a + bi$ and $w = c + di$

$|z| = \sqrt{a^2+b^2}$ and $|w| = \sqrt{c^2+d^2}$ $\therefore |z|^2 = a^2 + b^2$ and $|w|^2 = c^2 + d^2$

$|z+w| = \sqrt{(a+c)^2 + (b+d)^2}$ $\therefore |z+w|^2 = (a+c)^2 + (b+d)^2$

$|z-w| = \sqrt{(a-c)^2 + (b-d)^2}$ $\therefore |z-w|^2 = (a-c)^2 + (b-d)^2$

Now:

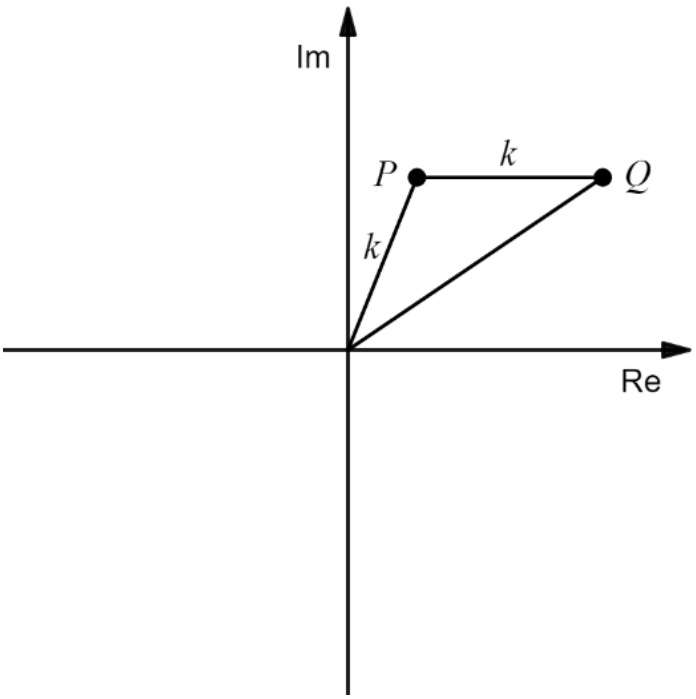
$$\begin{aligned} |z+w|^2 + |z-w|^2 &= (a+c)^2 + (b+d)^2 + (a-c)^2 + (b-d)^2 \\ &= a^2 + 2ac + c^2 + b^2 + 2bd + d^2 \\ &\quad + a^2 - 2ac + c^2 + b^2 - 2bd + d^2 \\ &= 2a^2 + 2c^2 + 2b^2 + 2d^2 \\ &= 2(a^2+b^2) + 2(c^2+d^2) \\ &= 2|z|^2 + 2|w|^2 \text{ as req.} \end{aligned}$$

50 a)

$$|z| = \sqrt{(k\cos\theta)^2 + (k\sin\theta)^2} = \sqrt{k^2(\cos^2\theta + \sin^2\theta)} = k$$

50 b)

Sketch z and $z + k$ on an Argand diagram:



Let $\arg(z+k)=\phi$ so $\angle OQP=\phi$ and since this is an isosceles triangle, $\angle POQ=\phi$

$$\therefore \arg(z+k)=\frac{1}{2}\arg(z)=\frac{1}{2}\theta$$

51 a)

$$\begin{aligned}\frac{z}{w}&=\frac{2+2\sqrt{3}i}{1+i}\times\frac{1-i}{1-i}=\frac{2+2\sqrt{3}i-2i+2\sqrt{3}}{1+1}\\&=(1+\sqrt{3})+(-1+\sqrt{3})i\end{aligned}$$

51 b)

$|z|=\sqrt{2^2+(2\sqrt{3})^2}=4$ and $\arg z=\arctan\left(\frac{2\sqrt{3}}{2}\right)=\frac{\pi}{3}$

$$z=4\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)$$

51 c)

$|w|=\sqrt{1^2+1^2}=\sqrt{2}$ and $\arg w=\arctan(1)=\frac{\pi}{4}$

$$w=\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)$$

51 d)

$$\begin{aligned}\frac{z}{w}&=\frac{4}{\sqrt{2}}\left(\cos\left(\frac{\pi}{3}-\frac{\pi}{4}\right)+i\sin\left(\frac{\pi}{3}-\frac{\pi}{4}\right)\right)\\&=2\sqrt{2}\left(\cos\frac{\pi}{12}+i\sin\frac{\pi}{12}\right)\end{aligned}$$

51 e)

$$(1+\sqrt{3})+(-1+\sqrt{3})i = 2\sqrt{2}\left(\cos\frac{\pi}{12}+i\sin\frac{\pi}{12}\right)$$

Comparing imaginary parts:

$$-1+\sqrt{3} = 2\sqrt{2}\sin\frac{\pi}{12}$$

$$\therefore \sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} \text{ as req.}$$

52 a)

$$w = 4\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right) = 2\sqrt{2}+2\sqrt{2}i$$

52 b)

$$\begin{aligned}\frac{z}{w} &= \frac{4\sqrt{3}-4i}{2\sqrt{2}+2\sqrt{2}i}\times\frac{2\sqrt{2}-2\sqrt{2}i}{2\sqrt{2}-2\sqrt{2}i}\\ &= \frac{8\sqrt{6}-8\sqrt{2}i-8\sqrt{2}-8\sqrt{6}i}{8+8}\\ &= \left(\frac{\sqrt{6}}{2}-\frac{\sqrt{2}}{2}\right)-\left(\frac{\sqrt{6}}{2}+\frac{\sqrt{2}}{2}\right)i\end{aligned}$$

52 c)

$|z| = \sqrt{(4\sqrt{3})^2+4^2} = 8$ and $\arg z = \arctan\left(\frac{4}{4\sqrt{3}}\right) = -\frac{\pi}{6}$

$$z = 8\left(\cos\left(-\frac{\pi}{6}\right)+i\sin\left(-\frac{\pi}{6}\right)\right)$$

52 d)

$8\div 4 = 2$ and $(-\frac{\pi}{6})-\frac{\pi}{4} = -\frac{5\pi}{12}$

$$\frac{z}{w} = 2\left(\cos\left(-\frac{5\pi}{12}\right)+i\sin\left(-\frac{5\pi}{12}\right)\right)$$

52 e)

$$\left(\frac{\sqrt{6}}{2}-\frac{\sqrt{2}}{2}\right)-\left(\frac{\sqrt{6}}{2}+\frac{\sqrt{2}}{2}\right)i = 2\left(\cos\left(-\frac{5\pi}{12}\right)+i\sin\left(-\frac{5\pi}{12}\right)\right)$$

Comparing real parts:

$$\frac{\sqrt{6}}{2}-\frac{\sqrt{2}}{2} = 2\cos\left(-\frac{5\pi}{12}\right)$$

$$\therefore \cos\left(-\frac{5\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4} \text{ as req.}$$

53 a)

$$\arg(wz) = \frac{\pi}{3} + \frac{\pi}{6} = \frac{\pi}{2}$$
$$\arg((wz)^2) = 2 \times \frac{\pi}{3} + 2 \times \frac{\pi}{6} = \pi$$
$$\arg((wz)^3) = 3 \times \frac{\pi}{3} + 3 \times \frac{\pi}{6} = \frac{3\pi}{2} \therefore \arg((wz)^3) = -\frac{\pi}{2}$$
$$\arg((wz)^4) = 4 \times \frac{\pi}{3} + 4 \times \frac{\pi}{6} = 2\pi \therefore \arg((wz)^4) = 0$$

$\therefore n = 4$ and $|(wz)^4| = 2^4 \times 5^4 = 10000$

53 b)

$$\arg(wz) = \frac{\pi}{6} + \frac{\pi}{2} = \frac{2\pi}{3}$$
$$\arg((wz)^2) = 2 \times \frac{\pi}{6} + 2 \times \frac{\pi}{2} = \frac{4\pi}{3} \therefore \arg((wz)^2) = -\frac{2\pi}{3}$$
$$\arg((wz)^3) = 3 \times \frac{\pi}{6} + 3 \times \frac{\pi}{2} = 2\pi \therefore \arg((wz)^3) = 0$$

$\therefore n = 3$ and $|(wz)^3| = 3^3 \times 8^3 = 13824$

53 c)

$$\arg(wz) = \frac{\pi}{4} - \frac{\pi}{12} = \frac{\pi}{6}$$
$$\arg((wz)^2) = 2 \times \frac{\pi}{4} - 2 \times \frac{\pi}{12} = \frac{\pi}{3}$$
$$\arg((wz)^3) = 3 \times \frac{\pi}{4} - 3 \times \frac{\pi}{12} = \frac{\pi}{2}$$
$$\arg((wz)^4) = 4 \times \frac{\pi}{4} - 4 \times \frac{\pi}{12} = \frac{2\pi}{3}$$

Continuing this gives $\frac{5\pi}{6}$, π, $\frac{7\pi}{6}$, $\frac{4\pi}{3}$, $\frac{3\pi}{2}$, $\frac{5\pi}{3}$, $\frac{11\pi}{6}$

$$\arg((wz)^{12}) = 12 \times \frac{\pi}{4} - 12 \times \frac{\pi}{12} = 2\pi \therefore \arg((wz)^{12}) = 0$$

$\therefore n = 12$ and $|(wz)^{12}| = \sqrt{2}^{12} \times 3^{12} = 34\,012\,224$

54)

$$\frac{1+\sqrt{3}i}{2-2i} \times \frac{2+2i}{2+2i} = \frac{(2-2\sqrt{3}) + (2+2\sqrt{3})i}{8}$$
$$= \frac{1-\sqrt{3}}{4} + \frac{1+\sqrt{3}}{4}i$$
$$\frac{1-\sqrt{3}}{4} + \frac{1+\sqrt{3}}{4}i = \frac{\sqrt{2}}{2}\left(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right)$$

Consider $\left(\frac{\sqrt{2}}{2}\left(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right)\right)^n$

Real means that the angle must be a multiple of π:

$$\therefore n = 12$$